



Classification

Prof. Gilad Gressel
Amrita School of Engineering

Introduction

- Difference between classification vs regression
- Binary classification
- Multi-class classification
- Multi-label classification

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Classification vs Regression

- Regression - predict a real valued number from a range of possibilities
- Classification - predict a class from a given set of classes



Binary Classification

- Examples
 - Dog or a cat?
 - Cancer or Benign?
 - Malicious or legitimate?
 - Buy or Sell?
- Can we do it?
 - Yes very well

Positive and Negative Classes



- In binary classification the positive class is always the class we are interested in detecting
- Malware detection:
 - Malware: 1, positive
 - Legitimate software: 0, negative
- Detecting Cancer
 - Cancerous tumor: 1, positive
 - Benign tumor: 0, negative

Multi-Class Classification



- Red, Orange, Yellow, or Blue?
- Husky, Beagle, Retriever, or Shepard?
- Sarcastic, Funny, Happy, Sad, or Bored?
- Nose, Eyes, Ears, or Hair?
- Can we do it?
 - Requires some work, but it's doable

Multi-Label Classification



- Harry Potter is: [boy, wizard, lives with muggles, scarred, “the chosen one”, plays quidditch]
- The news item is : [political, American, short, poorly written, satire]
- The food is : [Spicey, Indian, South-Indian]
- Can we do it?
 - Harder to design models to do this.

Conclusion

- Classification is discrete
- It's literally black and white
- Multi-class classification is common
- Multi-label classification (the tagging problem) is harder.

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Metrics

Prof. Gilad Gressel
Amrita School of Engineering

Introduction

- Measuring performance with metrics
- What kinds of things do we want to measure
- How do we measure them

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Metrics measure performance

- Definitions:
 - Performance : "The action or process of performing a task or function."
 - Metrics : "A system or standard of measurement."
- Why we care:
 - We need a quantifiable way to measure our performance.
 - How do we know if we are doing well on a certain task?

ML Tasks

- Supervised
 - Regression
 - Classification
- Unsupervised
 - Clustering
 - Dimensionality reduction

Regression Tasks



- Prediction on continuous functions
- How much will the new house cost? (from 0 to infinity)
- What will the temperature be tomorrow? (from 0 to 200)
- How many rooms do I need available for the busy season in December?
- What year was a song produced?
- How old is a person?

Classification Tasks

- Binary Classification
 - Is it a dog or cat?
 - Is it cancer or benign?
- Multiclass Classification
 - What kind of tumor is it? (from 13 options)
 - What malware family does it belong to? (from 5 options)

Things we want to measure in a model



- Correctness of the model
 - Accuracy, Precision, Recall, F1-score, Mean Squared Error, R2
- Speed of the model
 - Time to make prediction – inference
 - Time to train the model – training
- Resource Consumption
 - Ram, disk space, cpu speed, gpu cores
 - Dollars/prediction or dollars/training
- Data
 - Amount of data required to build your model
 - Cost to acquire the data

Measuring Correctness

- Classification (correctness) metrics rely on the two types of errors you can make
 - False Positives "This is a DOG!"
 - But..it was a cat
 - False Negatives "You are healthy!"
 - But.. you have cancer
- Regression (correctness) metrics utilize distances
 - We said 1900 but the answer is actually 1880
 - How "wrong" is that?

Conclusion

- We need a way to measure the performance of our model
- There are many different aspects
 - money, speed, correctness
- In this course we almost always focus on correctness

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Accuracy and Confusion

Prof. Gilad Gressel
Amrita School of Engineering

Introduction

- Metrics measure performance
- Accuracy
- Confusion matrix
- The problems with accuracy

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Accuracy



- Image we have a cat detection model and it has accuracy of 95%
- 95% of the time, the model is correct
- Accuracy equation:
 - $$\frac{TruePositives + TrueNegatives}{TruePositives + TrueNegatives + FalsePositives + FalseNegatives}$$

Accuracy for Cats n' Dogs



- We have a 100 pictures, 50 are dogs, 50 are cats
- Model found, 49 cats, 51 dogs
 - We said 49 things were cats, 47 were actually cats = 47 True Positives (TP)
 - But... 2 things we said were cats, were not = 2 False Positives (FP)
 - We said 51 things were dogs, 48 were actually dogs = 48 True Negatives (TN)
 - But... 3 things we said were dogs, were cats = 3 False Negatives (FN)
- $47(\text{TP}) + 48(\text{TN}) = 95$
- $47(\text{TP}) + 48(\text{TN}) + 3(\text{FN}) + 2(\text{FP}) = 100$
- $95/100 = 95\%$ accuracy



Confusion Matrix

		ACTUAL	ACTUAL
		Cat	Dog
PRED	Cat		
PRED	Dog		



Confusion Matrix

		ACTUAL	
		Cat	Dog
PRED	Cat	47 TP	2 FP
PRED	Dog	3 FN	48 TP

- 95% accuracy of our cat detector seems OK
- The rows are *predictions*, and the columns *actual*
- **Scikit-learn does it opposite!**
- That's why it's called the "confusion"

Accuracy is bad for imbalanced data



- 95% Accurate cancer model says:
- Predictions: "7 people have cancer, and 93 do not"
- Truth:
 - "We found 6 people who had cancer": TP=6
 - "But we said 1 person had cancer, who did not": FP=1
 - "We found 89 people who did not have cancer" TN=89
 - "We told 4 people they did not have cancer – but they did.": FN = 4
- TruePos 6 + True Neg 89 = 95
- TruePos 6 + TrueNeg 89 + FalseNeg 4 + FalsePos 1 = 100
- $95/100 = 95\%$ accuracy!

Cancer Confusion Matrix

		ACTUAL	ACTUAL
		1	0
PRED	1	6 TP	1 FP
PRED	0	4 FN	89 TN

- 95% accuracy is hiding the fact that we sent 4 people with cancer home!
- Why does accuracy fail?
 - The classes are imbalanced
 - One case *matters* a lot more

Conclusion

- Accuracy measures overall performance without differentiating between types of errors
- The confusion matrix shows us all 4 types of predictions
- Accuracy is not good for imbalanced datasets, or when one type of prediction is more important

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Classification Metrics

Prof. Gilad Gressel
Amrita School of Engineering

Introduction

- Confusion Matrix
- Precision
- Recall
- F1-score



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Confusion Matrix

		ACTUAL	ACTUAL
		1	0
PRED	1	6 TP	1 FP
PRED	0	4 FN	89 TN

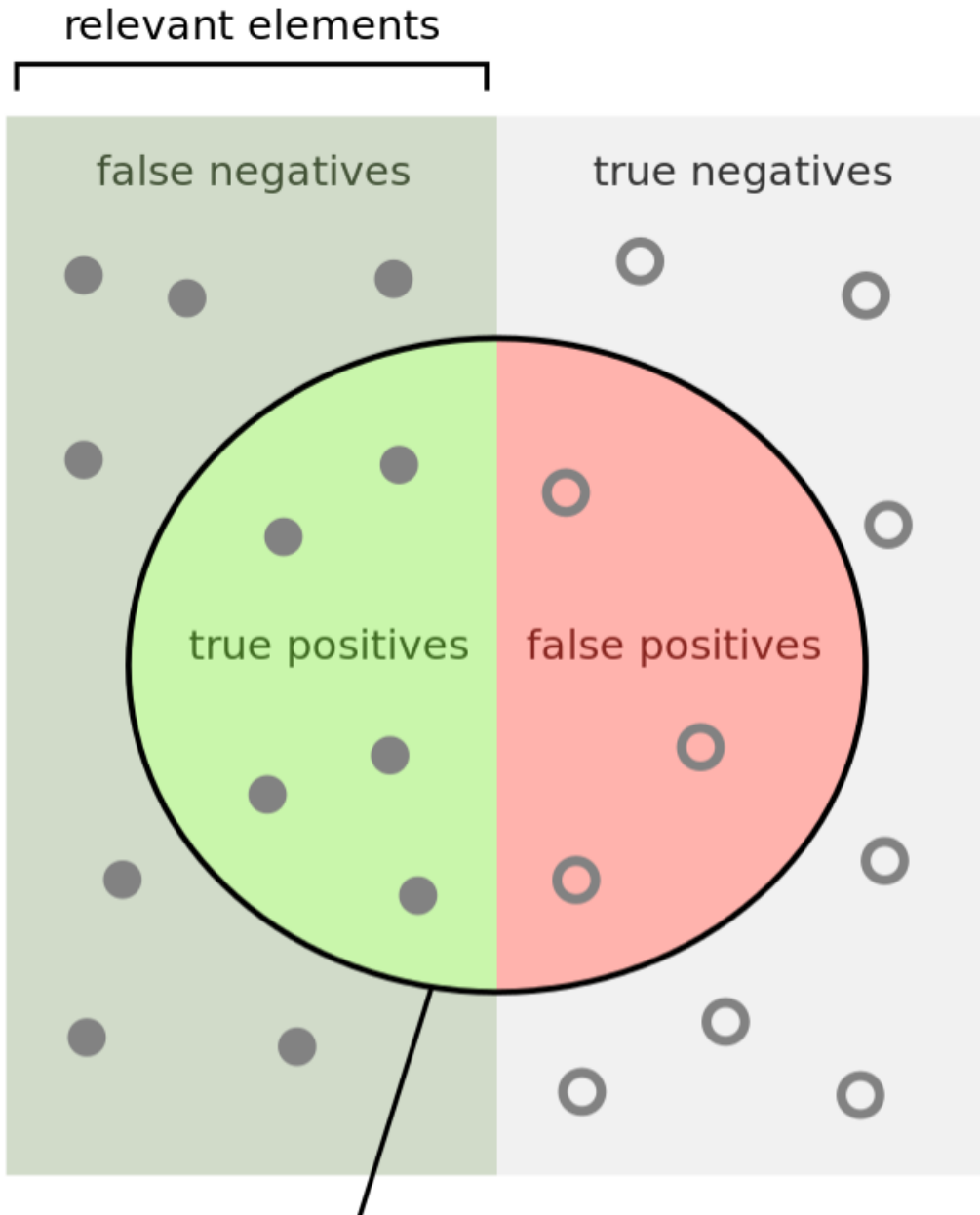
- The confusion matrix shows us 4 key components of a models performance
- It shows us the 4 different types of errors and positive predictions
- We draw many valuable statistics from it

https://en.wikipedia.org/wiki/Confusion_matrix

		True condition			
Total population		Condition positive	Condition negative	Prevalence = $\frac{\Sigma \text{Condition positive}}{\Sigma \text{Total population}}$	Accuracy (ACC) = $\frac{\Sigma \text{True positive} + \Sigma \text{True negative}}{\Sigma \text{Total population}}$
Predicted condition	Predicted condition positive	True positive	False positive, Type I error	Positive predictive value (PPV), Precision = $\frac{\Sigma \text{True positive}}{\Sigma \text{Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{False positive}}{\Sigma \text{Predicted condition positive}}$
	Predicted condition negative	False negative, Type II error	True negative	False omission rate (FOR) = $\frac{\Sigma \text{False negative}}{\Sigma \text{Predicted condition negative}}$	Negative predictive value (NPV) = $\frac{\Sigma \text{True negative}}{\Sigma \text{Predicted condition negative}}$
		True positive rate (TPR), Recall, Sensitivity, probability of detection, Power = $\frac{\Sigma \text{True positive}}{\Sigma \text{Condition positive}}$	False positive rate (FPR), Fall-out, probability of false alarm = $\frac{\Sigma \text{False positive}}{\Sigma \text{Condition negative}}$	Positive likelihood ratio (LR+) = $\frac{\text{TPR}}{\text{FPR}}$	Diagnostic odds ratio (DOR) = $\frac{\text{LR+}}{\text{LR-}}$ F ₁ score = $2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$
		False negative rate (FNR), Miss rate = $\frac{\Sigma \text{False negative}}{\Sigma \text{Condition positive}}$	Specificity (SPC), Selectivity, True negative rate (TNR) = $\frac{\Sigma \text{True negative}}{\Sigma \text{Condition negative}}$	Negative likelihood ratio (LR-) = $\frac{\text{FNR}}{\text{TNR}}$	

Precision and Recall

- Precision
 - If I said 7 people have cancer – how many actually did?
 - $6/7$
 - $TP / TP + FP$
 - Precision is “from what I said, how much was I correct about?”
- Recall
 - How many of the cancers did I find? (out of 10)
 - $6/10$
 - $TP / TP + FN$
 - Recall is: “how many did I get?”
- It's always about the **positive** class!



How many selected items are relevant?

$$\text{Precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}}$$

How many relevant items are selected?

$$\text{Recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$$

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<https://commons.wikimedia.org/w/index.php?curid=36926283>

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Confusion Matrix

		ACTUAL	ACTUAL
		1	0
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- Recall is the first column (actual truth!)
- Precision is the first row (Predicted truth!)

F-Score



- $F_{\beta} = (1 + \beta^2) \cdot \frac{\text{precision} \cdot \text{recall}}{(\beta^2 \cdot \text{precision}) + \text{recall}}$
- F1 resolves to: $2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$
 - Harmonic Mean – gives equal importance to precision and recall
 - Equals 0, if precision or recall are 0
 - Ill-defined if either precision and recall are 0.
- Beta > 1, then recall matters more
- Beta < 1, precision matters more.

Conclusion

- Confusion matrix gives us many metrics
- Precision, Recall, and F1-score summarize it well



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Choosing a Metric

Prof. Gilad Gressel
Amrita School of Engineering

Introduction

- Choosing a good metric is *very* important
- Using more than one is a good idea

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Choosing Good Metrics is Paramount

- The importance of choosing a metric(s) that matches your needs cannot be overstated
- All our decisions are based on metrics
- Bad metrics lead to bad decisions

How do you understand and choose a metric?



- There is no rule
- There is no “right way”
- There is only your own intelligence
- Try to get a “feeling” for what the metric is measuring by examining how the math plays out with small simple examples
- Read what others have used (and their justifications for why)
- Think about what is the most important aspect of your model you want to *measure* (what can you *not* afford to mess up?)



Beware Using a Single Metric

- Using a single metric will give you a single viewpoint
- This is very tempting and perhaps a good way to get started
- But problems arise quickly
 - Like paying teachers on how well students perform on tests
 - Optimizing youtube recommendations to keep users watching

Slate of Metrics

- Ideally you can have a number of metrics and you use them together
- Yes it's messier
- Yes it's better



Conclusion

- Choosing a metric is perhaps the most important thing you will do
- There are many metrics to choose from (we only looked at a few)
- You should learn the base math behind metrics.

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Lab Solution Practice with Metrics

Prof. Gilad Gressel
Amrita School of Engineering

Introduction

- Look at breast cancer data set
- Try a bunch of metrics
- Plot some matrices

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Conclusion

- Looked at different metrics
- Got practice plotting confusion matrices
- Took a gander at the classification report

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